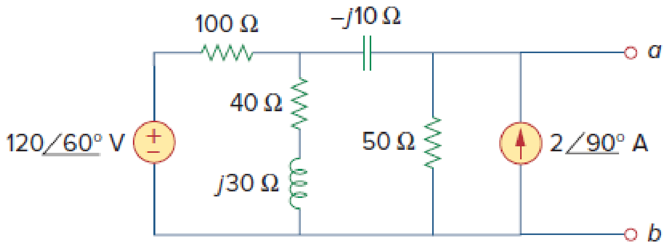
Chapter 11, Problem 21P

Problem

Assuming that the load impedance is to be purely resistive, what load should be connected to terminals *a-b* of the circuits in Fig. 11.52 so that the maximum power is transferred to the load?

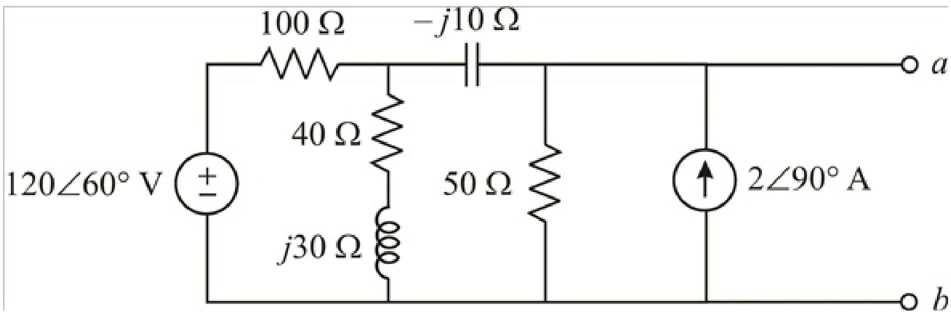
Figure 11.52:



Step-by-step solution

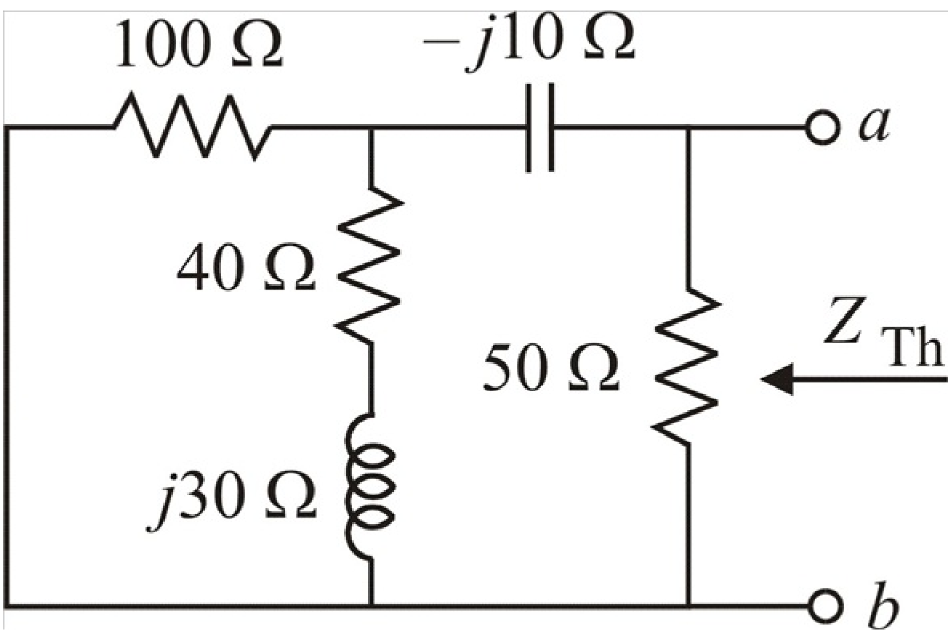
Step 1 of 4

The given circuit is,



Step 2 of 4

Short circuit the voltage source and open circuit the current source to find the Thevenin equivalent resistance across the terminals a-b,



Step 3 of 4

$$Z_{Th} = 50 \| \left[-j10 + \left(100 \| (40 + j30) \right) \right]$$

$$Z_{Th} = 50 \| \left[-j10 + \left(\frac{100(40 + j30)}{100 + 40 + j30} \right) \right]$$

$$Z_{Th} = 50 \| \left[-j10 + \frac{4000 + j3000}{140 + j30} \right]$$

$$Z_{Th} = 50 \| \left[\frac{(-j10)(140 + j30) + 4000 + j3000}{140 + j30} \right]$$

$$Z_{Th} = 50 \| \left[\frac{300 - j1400 + 4000 + j3000}{140 + j30} \right]$$

$$Z_{Th} = 50 \| \left(\frac{4300 + j1600}{140 + j30} \right)$$

Step 4 of 4

$$Z_{Th} = \frac{50\left(\frac{4300 + j1600}{140 + j30} \right)}{50 + \frac{4300 + j1600}{140 + j30}}$$

$$Z_{Th} = \frac{215000 + j80000}{7000 + j1500 + 4300 + j1600}$$

$$Z_{Th} = \frac{215000 + j80000}{11300 + j3100}$$

$$Z_{Th} = \frac{2150 + j800}{113 + j31}$$

$$Z_{Th} = \frac{2294.01\angle 20.41^\circ}{117.175\angle 15.34^\circ}$$

$$Z_{Th} = 19.577\angle 5.07^\circ$$

$$|Z_{Th}| = 19.58\ \Omega$$

Since the load is purely resistive, the condition for maximum power transfer is,

$$R_L = Z_{Th}$$

$$\therefore \boxed{R_L = 19.58\Omega}$$